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IS216 (G5/G6/G7) - Group Project - Proposal

IS216 Section	Group Number (eLearn)	Group Name
G5	8	SG Journey Planner

Group Members

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Problem Definition

(Write a paragraph that clearly articulates the problem being addressed, including the specific goals and any relevant constraints.)

Commuters in Singapore often need to check separate services for public-transport routes, live bus arrival times, and weather conditions before travelling. This makes it hard to decide when to leave and which journey option is most suitable, especially when rain can change walking conditions or make unsheltered stops less desirable. SG Journey Planner aims to combine route options, live public-transport information, weather data, and recent commuter reports in one application so users can make faster and more informed travel decisions.

Application Features

(List 4-5 key features/functionalities that the application will provide.)

1. **Smart Departure Recommender**
   The user enters an origin/destination or chooses a saved route. The application combines available journey options, live transport timings and current/near-term rain information. A simple rule-based score considers estimated travel time, waiting time, transfers and likely rain exposure, then recommends whether to leave now or wait briefly. The recommendation also shows the main reason so the result is understandable.

2. **Rain-Aware Route Finder**
   Using LTA location data, the application shows nearby bus and MRT stops on a map. Stop coordinates are matched to the nearest relevant weather area/forecast so users can see which nearby stops are currently affected by rain and choose a more comfortable boarding point.

3. **Route Comparison with Live Conditions**
   Users compare available journey options between the same origin and destination using estimated duration, transfers, walking distance, live transport information and current weather conditions.

4. **Weather-Triggered Trip Alerts/ Saved Routes**
   Users can save frequently used routes and their usual travel times. When the application is opened, it checks the relevant forecast and displays a warning when rain is expected around the saved travel time.

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5. **Community Reports**
   Users can submit quick, predefined reports for a bus stop or MRT station, such as crowded, sheltered/unsheltered, or wet conditions. Each report is timestamped and shown with its age, sorted newest first by default, and users can sort, filter and search reports by date or report type.

Members and Roles/Responsibilities

	Name	Roles/Responsibilities
1	Tan Guan Wei	**Smart Departure Recommender** 1. Build origin/destination search and journey request flow. 2. Retrieve available public-transport route options. 3. Retrieve live bus/transport timings where applicable. 4. Normalize route data such as duration, transfers, walking distance and waiting time for other features to consume. 5. Handle transport API loading, failure and no-route states.
2	Dillon	**Rain-Aware Route Finder** 1. Retrieve the user's/current selected location and nearby bus/MRT stops. 2. Build the nearby-stop map interface. 3. Retrieve current and near-term weather/rain information. 4. Match stop/route locations with the relevant weather area or forecast. 5. Provide reusable rain/weather information to the route comparison, recommender and saved-route features.
3	Amanda	**Route Comparison with Live Conditions** 1. Build the rule-based journey scoring system. 2. Compare options using travel time, waiting time, transfers, walking distance and rain exposure. 3. Generate Leave Now / Wait recommendations. 4. Produce understandable reasons for recommendations. 5. Build the route-comparison interface and test different journey/weather scenarios.
4	Woody	**Weather-Triggered Trip Alerts/ Saved Routes** 1. Implement adding/removing saved routes. 2. Store users' usual travel/departure times. 3. Load saved journeys when the application starts. 4. Check upcoming weather against saved travel times. 5. Display warnings or alerts when rain may affect an upcoming saved journey.
5	Anumitaa	**Community Reports & App Integration** 1. Build predefined reports such as crowded, wet and sheltered/unsheltered. 2. Associate reports with the correct bus stop or MRT station. 3. Timestamp reports and display their age. Reports are sorted newest first by default, and users can also sort, filter and search reports. 4. Display recent reports within relevant stop/station views. 5. Take ownership of cross-feature integration, end-to-end testing and ensuring the different modules work together.

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Discussion / Remarks / To-do list

Possible elements of engagement/ interactivity:

- Distinct visual indicators for status information (e.g. colour or icon states)
- Interactive elements such as tap-to-expand details
- Collapsible sections for each stop's details, so users see a summary first and expand only what they need